expanding forever, not having anywhere near enough matter in it to turn around the expansion or even slow down it very much.

Each of these possible universe types has a name and a particular geometry. The geometry isn't the external shape of the universe in the sense of it being a sphere or a cube or something. It's an internal property—something that can tell you how giant laser beams would behave while shooting across the cosmos on immense scales. (Because if you're going to measure a property of space, you may as well do it with giant laser beams.) A Big Crunch–fated universe is called a "closed" universe, because two parallel laser-cannon beams would eventually bend toward each other—it's the same kind of thing that happens to lines of longitude on a globe. What's happening in the cosmic case is that there's so much matter in a closed universe that *all of space* is curved inward. A perfectly balanced universe is "flat" because the beams would just stay parallel forever, in much the same way two parallel lines would stay parallel on a flat sheet of paper. A universe with way more expansion than gravity is called an "open" universe, and in that case, as you may have already guessed, the two laser beams would diverge from each other over time. The 2D-surface analog here is a saddle shape: try drawing parallel lines on a saddle (or, if you don't have a saddle handy, you can use a Pringles chip) and they'll get farther apart as they go. What these shapes represent is the "large-scale curvature" of the universe—the amount that space on the whole is distorted (or not) by the matter and energy within it.

The first thing all these possibilities have in common is that they all make sense, physically; they work well with Einstein's equations of gravity. The other is that for all of them, the present-day expansion is slowing down. At the time the supernova measurements were made, there was no reasonable physical mechanism to make a universe *speed up* in its expansion. It was just as weird as if you were throwing a ball up into the air and it slowed down a bit and then suddenly *shot off into space for no reason*. Exactly that weird except for the ENTIRE UNIVERSE.